Chapter 26 Back Matter Check

Contents

26 Supersymmetric Field Theories

Problems

1. Using the direct technique of Section 26.1 in the case of $N = 2$ supersymmetry, find the supersymmetry transformation rules of the massive field supermultiplet with just one Majorana spinor field and two complex scalars.

2. Calculate the component fields of a time-reversed superfield

$$T^{-1}S(x,\theta)T$$

in terms of the components of the superfield $S(x,\theta)$. What sort of superfield do we get by the time reversal of a left-chiral superfield? Of a linear superfield?

3. Consider the $N = 1$ supersymmetric theory of a single left-chiral superfield Φ. In superfield notation, list all the terms of dimensionality 5 involving Φ and/or Φ^* that might be added to the Lagrangian density.

4. Consider the theory of three left-chiral scalar superfields Φ_1, Φ_2, and Φ_3, with a conventional kinematic term, and with a superpotential

$$f(\Phi_1,\Phi_2,\Phi_3) = \Phi_1\Phi_3^2 + \Phi_2\big(\Phi_3^2 + a\big),$$

where a is a non-zero real constant. Show that this is a theory with spontaneously broken supersymmetry. Find the minimum value of the potential. Express the field of the goldstino in terms of the fermionic components of Φ_1, Φ_2, and Φ_3.

5. Find all the components of the current superfield for the action (26.6.9), in terms of the components of the left-chiral superfields Φ_n and derivatives of the superpotential f and the Kahler potential K.

6. Check that the supersymmetry current given by Eqs. (26.7.18) and (26.7.10) is related to the ω-component of the superfield (26.7.21) by Eq. (26.7.20).

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